Confirmatory Factor Analysis II

Introduction to Structural Equation Modeling

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Outline

Model Estimation

Model Fit

Model Complexity

Model Identification
- Under-Identified Models
- Over-Identified Models

Example

MODEL ESTIMATION

Model-Implied Statistics

Most statistical estimation algorithms operate by minimizing the difference between two key reference points:

1. The *model-implied* statistics/predictions/fitted values
   - The sufficient statistics implied by the structure of your model.
   - Predicted/fitted values produced by your model.
2. The *observed* statistics/values
   - The sufficient statistics calculated from the observed data.
   - The raw outcome values from your dataset.

The predictions/implied statistics produced by a good model must be simpler than the analogous quantities in the observed data.

- A model that exactly replicates the observed data is overfitting.
- The inferences from such models won't generalize to the population.

Model-Implied Statistics

You should already be familiar with this idea from OLS regression.

- The fitted values are the model implied statistics.

$$\hat{Y}_n = \hat{\beta}_0 + \sum_{p=1}^{P} \hat{\beta}_p X_{n,p}$$

- The raw outcome variable, Y, contains the observed values.
- Minimize the difference between $\hat{Y}$ and Y to estimate the model.

$$RSS = \sum_{n=1}^{N} \left(Y_n - \hat{Y}_n \right)^2$$

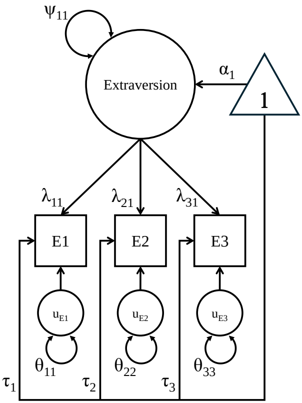
Hypothesized Model

We'll use this toy model to sketch out the intuition for maximum likelihood estimation of CFA models.

$$\alpha = \begin{bmatrix} \alpha_1 \end{bmatrix} \quad \Psi = \begin{bmatrix} \psi_{11} \end{bmatrix}$$

$$\tau = \begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \end{bmatrix} \quad \Lambda = \begin{bmatrix} \lambda_{11} \\ \lambda_{21} \\ \lambda_{31} \end{bmatrix}$$

$$\Theta = \begin{bmatrix} \theta_{11} & & \\ 0 & \theta_{22} & \\ 0 & 0 & \theta_{33} \end{bmatrix}$$



Parameters $\rightarrow$ Implied Statistics

The parameter matrices define the model-implied mean vector, $\hat{\mu}$, and covariance matrix, $\hat{\Sigma}$, via the following formulas.

$$\Sigma = \Lambda\Psi\Lambda^T + \Theta$$

$$\mu = \tau + \Lambda\alpha$$

By expanding these formulas, we can see how the model reproduces each mean, variance, and covariance.

$$\hat{\mu} = \begin{bmatrix} \tau_1 + \lambda_{11}\alpha_1 \\ \tau_2 + \lambda_{22}\alpha_1 \\ \tau_3 + \lambda_{33}\alpha_1 \end{bmatrix} \quad \hat{\Sigma} = \begin{bmatrix} \lambda_{11}\psi_{11}\lambda_{11} + \theta_{11} & & \\ \lambda_{11}\psi_{11}\lambda_{21} & \lambda_{21}\psi_{11}\lambda_{21} + \theta_{22} & \\ \lambda_{11}\psi_{11}\lambda_{31} & \lambda_{21}\psi_{11}\lambda_{31} & \lambda_{31}\psi_{11}\lambda_{31} + \theta_{33} \end{bmatrix}$$

Model-Implied Statistics for Estimation

The estimating algorithm chooses values for α, τ, Ψ, and Λ that minimize the differences between the model-implied statistics, $\{\hat{\mu}, \hat{\Sigma}\}$, and the sufficient statistics calculated from the observed data, $\{\bar{\mathbf{Y}}, \mathbf{S}\}$.

$$\hat{\mu} = \begin{bmatrix} \tau_1 + \lambda_{11}\alpha_1 \\ \tau_2 + \lambda_{22}\alpha_1 \\ \tau_3 + \lambda_{33}\alpha_1 \end{bmatrix} \quad \hat{\Sigma} = \begin{bmatrix} \lambda_{11}\psi_{11}\lambda_{11} + \theta_{11} & & \\ \lambda_{11}\psi_{11}\lambda_{21} & \lambda_{21}\psi_{11}\lambda_{21} + \theta_{22} & \\ \lambda_{11}\psi_{11}\lambda_{31} & \lambda_{21}\psi_{11}\lambda_{31} & \lambda_{31}\psi_{11}\lambda_{31} + \theta_{33} \end{bmatrix}$$

$$\bar{\mathbf{Y}} = \begin{bmatrix} \bar{y}_1 \\ \bar{y}_2 \\ \bar{y}_3 \end{bmatrix} \quad \mathbf{S} = \begin{bmatrix} var(y_1) & & \\ cov(y_2, y_1) & var(y_2) & \\ cov(y_3, y_1) & cov(y_3, y_2) & var(y_3) \end{bmatrix}$$

Optimization: Intuition

We can formulate the familiar OLS objective as an abstract optimization problem at follows.

$$f(\beta_0, \beta_1, \ldots, \beta_P) = \underset{\beta_0, \beta_1, \ldots, \beta_P}{\arg\min} \left\| Y - \hat{Y} \right\|$$

Applying the same idea to our CFA optimization problem, we get the following formulation.

$$f(\Lambda, \Psi) = \underset{\Lambda, \Psi}{\arg\min} \left\| \text{Cov}(\mathbf{Y}) - \hat{\Sigma} \right\|$$

$$f(\tau, \alpha) = \underset{\tau, \alpha}{\arg\min} \left\| \bar{\mathbf{Y}} - \hat{\mu} \right\|$$

Optimization: Objective Function

In OLS regression, we minimize the residual sum of squares (RSS).

$$RSS = \sum_{n=1}^{N} \left(Y_n - \hat{Y}_n \right)^2$$

In CFA, we can estimate the model by minimizing the *model deviance*.

$$D_M = -2 \left[\ell \left(\hat{\Sigma}, \hat{\mu} \right) - \ell \left(\mathbf{S}, \bar{\mathbf{Y}} \right) \right]$$

Where, $\ell(\cdot)$ denotes the *loglikelihood* function.

- $\ell \left(\hat{\Sigma}, \hat{\mu} \right)$ returns the loglikelihood value for our estimated model.
- $\ell \left(\mathbf{S}, \bar{\mathbf{Y}} \right)$ returns the loglikelihood value for the *saturated model*.

Optimization: Loglikelihood Functions

The loglikelihood function is a model-based distance metric.

- Derived from the assumed data distribution (i.e., multivariate normal)
- Quantifies the ”chances” of your estimated model having generated the observed data

$$\ell\left(\hat{\Sigma}, \hat{\mu}\right) = -\frac{K}{2}\ln(2\pi) - \frac{1}{2}\ln|\hat{\Sigma}| - \frac{1}{2}\sum_{n=1}^{N}\left(Y_n - \hat{\mu}\right)^T \hat{\Sigma}^{-1}\left(Y_n - \hat{\mu}\right)$$

Say we fit two models, M_a and M_b, to the same dataset. If $\ell\left(M_a\right)$ is closer to zero, we know that M_a represents the data better than M_b does.

- M_a's implied statistics are closer to the observed sufficient statistics.
- M_a's parameters are more plausible, given the data.

MODEL FIT

Model Fit: Intuition

We want a plausible, parsimonious, and useful model.

1. Our model should make theoretical sense.
2. Our model's implied statistics should closely align with the observed sufficient statistics.
3. Our model should be no more complex than necessary.

We can use *model fit statistics* to evaluate the second and third criteria.

- Model fit statistics transform $\hat{D}_M$ into interpretable, comparable, scale-free distance measures.
- These indices give us practical tools for evaluating how well our model represents the data.

Different Ways to Quantify Fit

When evaluating model fit, we need to consider classes of model.

1. Our estimated model.
2. The *independence model*
   - The worst possible model
   - All item associations are fixed to zero
3. The *saturated model*
   - The ”best” possible model
   - All possible associations are estimated
   - Perfectly reproduces the observed sufficient statistics

Independence Model

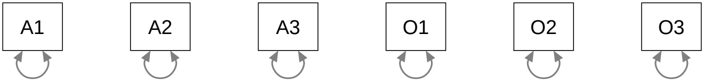
```
indMod <- '
A1 ~~ A1
A2 ~~ A2
A3 ~~ A3
O1 ~~ O1
O2 ~~ O2
O3 ~~ O3
'

indOut <- cfa(indMod, data = bfi)
```

Independence Model

```
partSummary(indOut, 1:4)

lavaan 0.6-19 ended normally after 20 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                         6

                                                  Used       Total
  Number of observations                          2696        2800

Model Test User Model:

  Test statistic                              2053.922
  Degrees of freedom                                15
  P-value (Chi-square)                           0.000
```

Independence Model

```
lavInspect(indOut, "cov.ov")

      A1    A2    A3    O1    O2    O3
A1 1.984
A2 0.000 1.377
A3 0.000 0.000 1.708
O1 0.000 0.000 0.000 1.271
O2 0.000 0.000 0.000 0.000 2.431
O3 0.000 0.000 0.000 0.000 0.000 1.485
```



Saturated Model

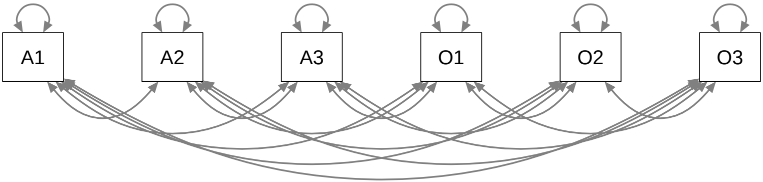
```
satMod <- '
A1 ~~ A1
A2 ~~ A1 + A2
A3 ~~ A1 + A2 + A3
O1 ~~ A1 + A2 + A3 + O1
O2 ~~ A1 + A2 + A3 + O1 + O2
O3 ~~ A1 + A2 + A3 + O1 + O2 + O3
'

satOut <- cfa(satMod, data = bfi)
```

Saturated Model

```
partSummary(satOut, 1:4)

lavaan 0.6-19 ended normally after 36 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        21

                                                  Used       Total
  Number of observations                          2696        2800

Model Test User Model:

  Test statistic                                 0.000
  Degrees of freedom                                 0
```

Saturated Model

```
lavInspect(satOut, "cov.ov")

       A1     A2     A3     O1     O2     O3
A1  1.984
A2 -0.567  1.377
A3 -0.493  0.751  1.708
O1  0.013  0.174  0.213  1.271
O2  0.172  0.024 -0.004 -0.382  2.431
O3 -0.098  0.234  0.357  0.544 -0.514  1.485
```



Measurement Structure

We don't estimate any measurement structure in the independence model or in the saturated model.

```
lavInspect(indOut, "est")$lambda


A1
A2
A3
O1
O2
O3

lavInspect(indOut, "est")$psi

<0 x 0 matrix>
```

```
lavInspect(satOut, "est")$lambda


A1
A2
A3
O1
O2
O3

lavInspect(satOut, "est")$psi

<0 x 0 matrix>
```

Different Ways to Quantify Fit

We can conceptualize model fit comparisons in two ways.

1. *Absolute Fit*
    - How well does the model reproduce the observed sufficient statistics?
    - How close is our model to the best possible model?
    - Compare our model to the *saturated model*.

2. *Relative/Incremental/Comparative Fit*
    - How much better does our model perform than some poorer baseline model?
    - How much improvement do we achieve over the worst possible model?
    - Compare our model to the *indepenence model*.

χ^2 Statistic

The χ^2 statistic is the most fundamental model fit statistic.

- Absolute fit statistic
- Derived directly from the optimized value of the objective function
- Allows rigorous hypothesis tests
- Basis for most other fit indices

$$\chi^2 = D_M$$
$$\chi^2 = N \times F_{ML}$$

This statistic follows a χ^2 distribution with $df = Q - P$ (i.e., the same DF as the estimated model).

Example Model

```r
library(lavaan)
library(psych)

mod1 <- '
agree =~ A1 + A2 + A3
open  =~ O1 + O2 + O3
'

out1 <- cfa(mod1, data = bfi)
```

Example Model

```
partSummary(out1, 1:4)

lavaan 0.6-19 ended normally after 40 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        13

                                                  Used       Total
  Number of observations                          2696        2800

Model Test User Model:

  Test statistic                               120.283
  Degrees of freedom                                 8
  P-value (Chi-square)                           0.000
```

Problems with χ^2

The χ^2 statistic test for *exact fit*:

$$
\begin{aligned}
H_0 &: \hat{\Sigma} = \mathbf{S} \\
H_1 &: \hat{\Sigma} \neq \mathbf{S}
\end{aligned}
$$

This hypothesis is perfectly valid, but it isn't a very interesting or useful for real-world data analysis.

- The χ^2 will almost always reject the null hypothesis.
- With a large enough sample, the null hypothesis must be reject for any model other than the exact data generating model.
- We'll reject perfectly good models because of substantively trivial deviations from the population model.

We don't want a model that exactly replicates the sufficient statistics.

- The true data generating model is almost certainly too complex to have any practical utility.

Approximate Fit Indices

In practice, we won't use the χ^2 test to directly evaluate model fit. We'll use *approximate fit indices* (mostly) derived from the χ^2.

- We have many (maybe too many) options.
- **lavaan** 0.6-19 provides 28 approximate fit measures for a basic CFA.

We'll focus on three complementary and well-balanced indices.

- Root Mean Squared Error of Approximation (RMSEA)
- Comparative Fit Index (CFI)
- Standardized Root Mean Squared Residual (SRMR)

RMSEA

The RMSEA is an *absolute fit index* that evaluates deviations from *close fit*.

$$\text{RMSEA} = \sqrt{\max\left[\frac{\chi^2 - df}{N \times df},\ 0\right]}$$

CFI

The CFI is a normed *incremental fit index*.

$$\text{CFI} = 1 - \frac{\max\left[(\chi^2 - df),\ 0\right]}{\max\left[(\chi^2_0 - df_0),\ (\chi^2 - df),\ 0\right]}$$

Assuming the ordinary situation where $\chi^2 \geq df$, and the baseline model fits worse than our target model, we can simplify the CFI formula.

$$CFI = 1 - \frac{\chi^2 - df}{\chi^2_0 - df_0}$$

SRMR

$$\text{SRMR} = \sqrt{\frac{\sum_{i=1}^{M} \sum_{j=1}^{i} \left[\frac{s_{ij} - \hat{\sigma}_{ij}}{s_i s_j} \right]^2}{Q}}$$

Example: Target Model

We can use the `fitMeasures()` function to get model fit statistics.

```
fit1 <- fitMeasures(out1)

head(fit1, 22) |> round(3)

             npar              fmin             chisq                df
           13.000             0.022           120.283             8.000
           pvalue    baseline.chisq       baseline.df   baseline.pvalue
            0.000          2053.922            15.000             0.000
              cfi               tli              nnfi               rfi
            0.945             0.897             0.897             0.890
              nfi              pnfi               ifi               rni
            0.941             0.502             0.945             0.945
             logl unrestricted.logl               aic               bic
       -26114.771        -26054.629         52255.542         52332.236
           ntotal              bic2
         2696.000         52290.931
```

Example: Target Model

```r
tail(fit1, -22) |> round(3)
```

```
                 rmsea          rmsea.ci.lower          rmsea.ci.upper 
                 0.072                   0.061                   0.084 
        rmsea.ci.level            rmsea.pvalue          rmsea.close.h0 
                 0.900                   0.001                   0.050 
rmsea.notclose.pvalue      rmsea.notclose.h0                     rmr 
                 0.138                   0.080                   0.068 
            rmr_nomean                    srmr            srmr_bentler 
                 0.068                   0.039                   0.039 
   srmr_bentler_nomean                    crmr             crmr_nomean 
                 0.039                   0.046                   0.046 
            srmr_mplus       srmr_mplus_nomean                   cn_05 
                 0.039                   0.039                 348.577 
                 cn_01                     gfi                    agfi 
               451.298                   0.986                   0.963 
                  pgfi                     mfi                    ecvi 
                 0.376                   0.979                   0.054 
```

Example: Independence Model

```r
indFit <- fitMeasures(indOut) |> as.list()
fit1   <- as.list(fit1)

indFit$chisq

[1] 2053.922

fit1$baseline.chisq

[1] 2053.922

indFit$df

[1] 15

fit1$baseline.df

[1] 15
```

Example: Saturated Model

The saturated model has perfect fit because it perfectly reproduces the observed sufficient statistics.

```
fitMeasures(satOut, c("chisq", "df", "rmsea", "cfi", "srmr"))

chisq     df  rmsea    cfi   srmr
    0      0      0      1      0
```

Manually Calculating χ^2

```
llSat <- logLik(satOut) # saturated model loglikelihood
ll1   <- logLik(out1)   # target model loglikelihood

# Manually calculate the chi-squared stat
-2 * (ll1 - llSat) |> as.numeric()

[1] 120.2832

# Compare to lavaan's version
fit1$chisq

[1] 120.2832
```

Manually Calculating CFI

```r
chisq1 <- fit1$chisq
df1    <- fit1$df
chisq0 <- fit1$baseline.chisq
df0    <- fit1$baseline.df

# Manually calculate the CFI
1 - ((chisq1 - df1) / (chisq0 - df0))
```

```
[1] 0.9449301
```

```r
# Compare to lavaan's version
fit1$cfi
```

```
[1] 0.9449301
```

Manually Calculating RMSEA

```
n <- fit1$ntotal

# Manually calculate the RMSEA
sqrt((chisq1 - df1) / (n * df1))

[1] 0.07215267

# Compare to lavaan's version
fit1$rmsea

[1] 0.07215267
```

Manually Calculating SRMR

```r
M <- 6                 # No. of observed variables
Q <- M * (M + 1) / 2 # No. of unique elements in S

# Model-implied covariance matrix
sHat <- inspect(out1, "sigma")

# Observed covariance matrix
sObs <- inspect(out1, "sampstat")$cov

# Observed standard deviations
se   <- sqrt(diag(sObs))
```

Manually Calculating SRMR

```
# Sum of squared standardized residuals
x <- 0
for(i in 1:M) {
  for(j in 1:i) {
    x <- x + ( (sObs[i, j] - sHat[i, j]) / (se[i] * se[j]) )^2
  }
}

# Root mean of the above = SRMR
sqrt(x / Q) |> as.numeric()

[1] 0.03850419

# Compare to lavaan's version
fit1$srmr

[1] 0.03850419
```

MODEL COMPLEXITY

DF & Complexity

Parsimony

Model Identification

Identifying Constraints

We must fix some parameters to identify the model.

- For each construct, we need $df \geq 0$.
- If the construct has three or more indicators:
  - Fix one parameter in the covariance model.
  - Fix one parameter in the mean model.
- If the construct has two indicators:
  - Fix an additional parameter in the covariance model.
- If the construct has only one indicator:
  - Cannot define a latent factor.

Under-Identified: Unconstrained Model

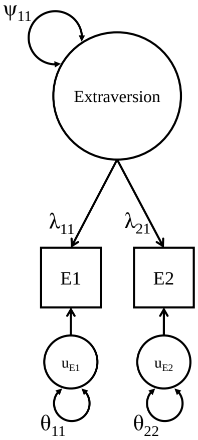
$$\Psi = \begin{bmatrix} \psi_{11} \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} \lambda_{11} \\ \lambda_{21} \end{bmatrix}$$

$$\Theta = \begin{bmatrix} \theta_{11} & \\ 0 & \theta_{22} \end{bmatrix}$$

$$Q = \frac{2(2+1)}{2} = 3$$

$$\begin{aligned} df &= Q - P \\ &= 3 - 5 = -2 \end{aligned}$$



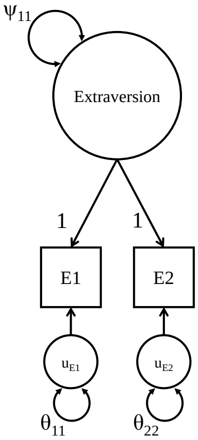
Under-Identified: Marker-Variable

$$\Psi = \begin{bmatrix} \psi_{11} \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Theta = \begin{bmatrix} \theta_{11} & \\ 0 & \theta_{22} \end{bmatrix}$$

$$\begin{aligned} df &= Q - P \\ &= 3 - 3 = 0 \end{aligned}$$



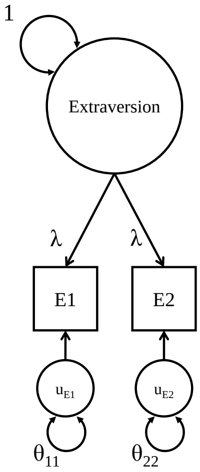
Under-Identified: Fixed-Factor

$$\Psi = \begin{bmatrix} 1 \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} \lambda \\ \lambda \end{bmatrix}$$

$$\Theta = \begin{bmatrix} \theta_{11} & \\ 0 & \theta_{22} \end{bmatrix}$$

$$\begin{aligned} df &= Q - P \\ &= 3 - 3 = 0 \end{aligned}$$



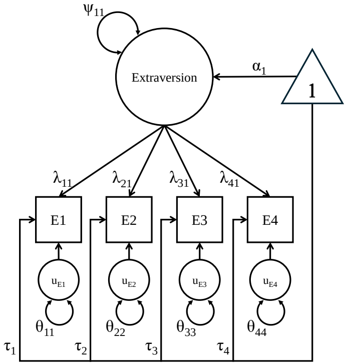
Over-Identified Models

What happens when we have four (or more) indicators?

With 4 indicators, our model contains 14 parameters:

- Four factor loadings
- Four residual variances
- Four item intercepts
- One latent mean
- One latent variance

$$Q = \frac{4(4+1)}{2} + 4 = 14$$
$$df = 14 - 14 = 0$$



Over-Identified Models

With 5 indicators, we have 17 model parameters:

- Five factor loadings
- Five residual variances
- Five item intercepts
- One latent mean
- One latent variance

$$Q = \frac{5(5+1)}{2} + 5 = 20$$
$$df = 20 - 17 = 3$$

With 6 indicators, we have 20 model parameters:

- Six factor loadings
- Six residual variances
- Six item intercepts
- One latent mean
- One latent variance

$$Q = \frac{6(6+1)}{2} + 6 = 27$$
$$df = 27 - 20 = 7$$

Over-Identified Models

With four indicators, we automatically have $df \geq 0$ without imposing any scaling constraints.

- Can we directly estimate the unconstrained model?

```
mod1 <- '
fun =~ Q77 + Q84 + Q170 + Q196
'

fit1 <- lavaan(mod1,
               data          = sapi,
               auto.var      = TRUE,
               meanstructure = TRUE,
               int.ov.free   = TRUE,
               int.lv.free   = TRUE)

Warning:  lavaan->lav_model_vcov():
  Could not compute standard errors!  The information matrix could not be
  inverted.  This may be a symptom that the model is not identified.
```

Hmmm...I guess not.

Over-Identified Models

```
partSummary(fit1, 1:4)

lavaan 0.6-19 ended normally after 13 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        14

                                                  Used       Total
  Number of observations                           970        1000

Model Test User Model:

  Test statistic                                58.017
  Degrees of freedom                                 0
```

Over-Identified Models

```
partSummary(fit1, 7:8)

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)
  fun =~
    Q77               0.824 NA
    Q84               0.578 NA
    Q170              0.479 NA
    Q196              0.619 NA

Intercepts:
                   Estimate  Std.Err  z-value  P(>|z|)
   .Q77               3.574 NA
   .Q84               3.232 NA
   .Q170              3.955 NA
   .Q196              3.803 NA
    fun               0.000 NA
```

Over-Identified Models

```
partSummary(fit1, 9)

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)
   .Q77               0.491 NA
   .Q84               0.760 NA
   .Q170              0.703 NA
   .Q196              0.370 NA
    fun               1.012 NA
```

Over-Identified Models

In the above example, the data contain enough information to define our model parameters, but that's not enough.

- We still need scaling constraints.
- The estimation algorithm needs an anchor point from which to extrapolate the relative values of the model parameters.
- Without any scaling constraints, an infinite number of parameter matrices will produce the same $\hat{\mu}$ and $\hat{\Sigma}$.
- An infinite number of solutions are equally good.

```r
## Fit a model to get some example parameters:
mod1 <- '
fun   =~ Q77 + Q84 + Q170 + Q196
liked =~ Q44 + Q63 + Q76  + Q98
'

fit1 <- cfa(mod1, data = sapi, effect.coding = TRUE)
```

Over-Identified Models

```
## Extract the estimated factor loadings and latent covariance matrix:
(lambda <- inspect(fit1, "est")$lambda)

       fun liked
Q77  1.254 0.000
Q84  0.954 0.000
Q170 0.795 0.000
Q196 0.997 0.000
Q44  0.000 0.764
Q63  0.000 1.155
Q76  0.000 1.133
Q98  0.000 0.949

(psi <- inspect(fit1, "est")$psi)

        fun liked
fun   0.399
liked 0.241 0.311
```

Over-Identified Models

```r
## Extract the estimated residual variances:
(theta <- inspect(fit1, "est")$theta)

       Q77   Q84  Q170  Q196   Q44   Q63   Q76   Q98
Q77  0.548
Q84  0.000 0.727
Q170 0.000 0.000 0.687
Q196 0.000 0.000 0.000 0.364
Q44  0.000 0.000 0.000 0.000 0.662
Q63  0.000 0.000 0.000 0.000 0.000 0.807
Q76  0.000 0.000 0.000 0.000 0.000 0.000 0.966
Q98  0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.469
```

Over-Identified Models

```r
## Manually compute the model-implied covariance matrix:
(sigma  <- lambda %*% psi %*% t(lambda) + theta)

       Q77   Q84  Q170  Q196   Q44   Q63   Q76   Q98
Q77  1.176
Q84  0.477 1.090
Q170 0.398 0.303 0.940
Q196 0.499 0.380 0.316 0.761
Q44  0.231 0.175 0.146 0.183 0.843
Q63  0.349 0.265 0.221 0.277 0.275 1.223
Q76  0.342 0.260 0.217 0.272 0.269 0.407 1.365
Q98  0.287 0.218 0.182 0.228 0.226 0.341 0.335 0.750
```

Over-Identified Models

```
## Randomly sample an arbitrary scaling factor:
(a <- runif(1, 1, 2))

[1] 1.524314

## Rescale all factor loadings by a factor of a:
(lambda2 <- lambda * a)

       fun liked
Q77  1.911 0.000
Q84  1.454 0.000
Q170 1.212 0.000
Q196 1.520 0.000
Q44  0.000 1.164
Q63  0.000 1.760
Q76  0.000 1.727
Q98  0.000 1.447
```

Over-Identified Models

```
## Rescale the latent covariance matrix by a factor of (1 / a^2):
(psi2 <- psi / a^2)

        fun liked
fun   0.172
liked 0.104 0.134

## Compute the model-implied covariance matrix using the rescaled parameters:
(sigma2  <- lambda2 %*% psi2 %*% t(lambda2) + theta)

       Q77   Q84  Q170  Q196   Q44   Q63   Q76   Q98
Q77  1.176
Q84  0.477 1.090
Q170 0.398 0.303 0.940
Q196 0.499 0.380 0.316 0.761
Q44  0.231 0.175 0.146 0.183 0.843
Q63  0.349 0.265 0.221 0.277 0.275 1.223
Q76  0.342 0.260 0.217 0.272 0.269 0.407 1.365
Q98  0.287 0.218 0.182 0.228 0.226 0.341 0.335 0.750
```

Over-Identified Models

```
## Compare the two model-implied covariance matrices:
all.equal(sigma, sigma2)

[1] TRUE

## Repeat the experiment 100 times:
out <- rep(NA, 100)
for(i in 1:100) {
  a <- runif(1, 1, 2)

  lambda2 <- lambda * a
  psi2    <- psi / a^2
  sigma2  <- lambda2 %*% psi2 %*% t(lambda2) + theta

  out[i] <- all.equal(sigma, sigma2)
}

## Always the same result?
all(out)

[1] TRUE
```

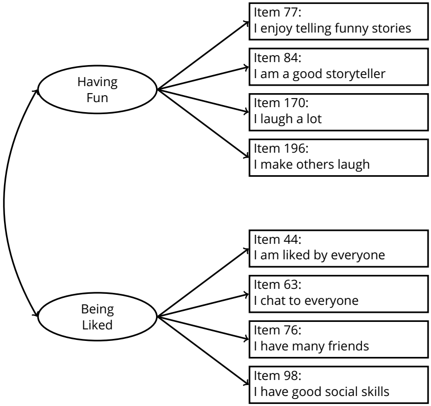
EXAMPLE

CFA of Extraversion Items

Suppose we hypothesize two distinct dimensions of extraversion underlying 8 of the SAPI items.

1. Having Fun
2. Being Liked by Others

We'll define our measurement model as the two-factor CFA shown to the right.



Example: Marker Variable

Load the SAPI data.

```
dataDir <- "data"
sapi <- read.table(here::here(dataDir, "sapi.txt"),
                   header = TRUE,
                   na.strings = "-999")
```

Specify the **lavaan** model syntax for the SAPI extraversion CFA.

```
mod1 <- '
fun   =~ Q77 + Q84 + Q170 + Q196
liked =~ Q44 + Q63 + Q76  + Q98
'
```

Use the `cfa()` function to estimate the model.

```
library(lavaan)
out1 <- cfa(mod1, data = sapi)
```

Example: Marker Variable

```
partSummary(out1, 1:4)

lavaan 0.6-19 ended normally after 30 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        17

                                                  Used       Total
  Number of observations                           959        1000

Model Test User Model:

  Test statistic                               130.193
  Degrees of freedom                                19
  P-value (Chi-square)                           0.000
```

Example: Marker Variable

```
partSummary(out1, 5:7)

Parameter Estimates:

  Standard errors                                  Standard
  Information                                      Expected
  Information saturated (h1) model               Structured

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)
  fun =~
    Q77               1.000
    Q84               0.761    0.051   14.902    0.000
    Q170              0.634    0.047   13.558    0.000
    Q196              0.795    0.046   17.381    0.000
  liked =~
    Q44               1.000
    Q63               1.512    0.147   10.278    0.000
    Q76               1.483    0.149    9.955    0.000
    Q98               1.243    0.119   10.462    0.000
```

Example: Marker Variable

```
partSummary(out1, 8:9)

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)
  fun ~~
    liked             0.231    0.025    9.234    0.000

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)
   .Q77               0.548    0.038   14.389    0.000
   .Q84               0.727    0.039   18.703    0.000
   .Q170              0.687    0.035   19.572    0.000
   .Q196              0.364    0.025   14.731    0.000
   .Q44               0.662    0.034   19.291    0.000
   .Q63               0.807    0.048   16.943    0.000
   .Q76               0.966    0.054   17.931    0.000
   .Q98               0.469    0.029   16.121    0.000
    fun               0.627    0.056   11.303    0.000
    liked             0.182    0.029    6.290    0.000
```

Example: Marker Variable

```r
inspect(out1, "r2")

  Q77   Q84  Q170  Q196   Q44   Q63   Q76   Q98
0.534 0.333 0.268 0.521 0.215 0.340 0.293 0.374

fitMeasures(out1) |> head(22) |> round(3)

               npar                fmin               chisq                  df
             17.000               0.068             130.193              19.000
             pvalue      baseline.chisq         baseline.df     baseline.pvalue
              0.000            1574.886              28.000               0.000
                cfi                 tli                nnfi                 rfi
              0.928               0.894               0.894               0.878
                nfi                pnfi                 ifi                 rni
              0.917               0.622               0.929               0.928
               logl  unrestricted.logl                 aic                 bic
         -10147.587          -10082.491           20329.175           20411.895
             ntotal                bic2
            959.000           20357.903
```

Example: Marker Variable

```
fitMeasures(out1) |> tail(-22) |> round(3)

               rmsea          rmsea.ci.lower          rmsea.ci.upper
               0.078                   0.066                   0.091
      rmsea.ci.level           rmsea.pvalue          rmsea.close.h0
               0.900                   0.000                   0.050
rmsea.notclose.pvalue     rmsea.notclose.h0                     rmr
               0.421                   0.080                   0.042
          rmr_nomean                    srmr            srmr_bentler
               0.042                   0.043                   0.043
  srmr_bentler_nomean                   crmr             crmr_nomean
               0.043                   0.049                   0.049
          srmr_mplus       srmr_mplus_nomean                   cn_05
               0.043                   0.043                 223.037
               cn_01                     gfi                    agfi
             267.582                   0.968                   0.939
                pgfi                     mfi                    ecvi
               0.511                   0.944                   0.171
```

Example: Fixed Factor

We only need to change one option to implement the fixed-factor method.

- The `std.lv = TRUE` option (i.e., standardized latent variables) applies the appropriate constraints.

```
out2 <- cfa(mod1, data = sapi, std.lv = TRUE)
```

Example: Fixed Factor

```
partSummary(out2, 1:4)

lavaan 0.6-19 ended normally after 17 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        17

                                                  Used       Total
  Number of observations                           959        1000

Model Test User Model:

  Test statistic                               130.193
  Degrees of freedom                                19
  P-value (Chi-square)                           0.000
```

Example: Fixed Factor

```
partSummary(out2, 5:7)

Parameter Estimates:

  Standard errors                                 Standard
  Information                                     Expected
  Information saturated (h1) model              Structured

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)
  fun =~
    Q77               0.792    0.035   22.606    0.000
    Q84               0.603    0.035   17.193    0.000
    Q170              0.502    0.033   15.180    0.000
    Q196              0.630    0.028   22.308    0.000
  liked =~
    Q44               0.426    0.034   12.580    0.000
    Q63               0.644    0.040   16.071    0.000
    Q76               0.632    0.043   14.845    0.000
    Q98               0.530    0.031   16.912    0.000
```

Example: Fixed Factor

```
partSummary(out2, 8:9)

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)
  fun ~~
    liked             0.683    0.033   20.483    0.000

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)
   .Q77               0.548    0.038   14.389    0.000
   .Q84               0.727    0.039   18.703    0.000
   .Q170              0.687    0.035   19.572    0.000
   .Q196              0.364    0.025   14.731    0.000
   .Q44               0.662    0.034   19.291    0.000
   .Q63               0.807    0.048   16.943    0.000
   .Q76               0.966    0.054   17.931    0.000
   .Q98               0.469    0.029   16.121    0.000
    fun               1.000
    liked             1.000
```

Example: Fixed Factor

```
inspect(out2, "r2")

  Q77   Q84  Q170  Q196   Q44   Q63   Q76   Q98
0.534 0.333 0.268 0.521 0.215 0.340 0.293 0.374

fitMeasures(out2) |> head(22) |> round(3)

             npar              fmin             chisq                df
           17.000             0.068           130.193            19.000
           pvalue    baseline.chisq       baseline.df   baseline.pvalue
            0.000          1574.886            28.000             0.000
              cfi               tli              nnfi               rfi
            0.928             0.894             0.894             0.878
              nfi              pnfi               ifi               rni
            0.917             0.622             0.929             0.928
             logl unrestricted.logl               aic               bic
       -10147.587        -10082.491         20329.175         20411.895
           ntotal              bic2
          959.000         20357.903
```

Example: Fixed Factor

```
fitMeasures(out2) |> tail(-22) |> round(3)

                rmsea          rmsea.ci.lower          rmsea.ci.upper
                0.078                   0.066                   0.091
       rmsea.ci.level            rmsea.pvalue          rmsea.close.h0
                0.900                   0.000                   0.050
rmsea.notclose.pvalue     rmsea.notclose.h0                     rmr
                0.421                   0.080                   0.042
           rmr_nomean                    srmr            srmr_bentler
                0.042                   0.043                   0.043
  srmr_bentler_nomean                    crmr             crmr_nomean
                0.043                   0.049                   0.049
           srmr_mplus       srmr_mplus_nomean                   cn_05
                0.043                   0.043                 223.037
                cn_01                     gfi                    agfi
              267.582                   0.968                   0.939
                 pgfi                     mfi                    ecvi
                0.511                   0.944                   0.171
```

Compare Results

```
inspect(out1, "est")$lambda - inspect(out2, "est")$lambda

       fun liked
Q77   0.208 0.000
Q84   0.158 0.000
Q170  0.132 0.000
Q196  0.165 0.000
Q44   0.000 0.574
Q63   0.000 0.868
Q76   0.000 0.851
Q98   0.000 0.713

inspect(out1, "est")$psi - inspect(out2, "est")$psi

         fun   liked
fun    -0.373
liked  -0.453 -0.818
```

Compare Results

```
all.equal(fitMeasures(out1), fitMeasures(out2))

[1] TRUE

inspect(out1, "r2") - inspect(out2, "r2")

 Q77  Q84 Q170 Q196  Q44  Q63  Q76  Q98
   0    0    0    0    0    0    0    0

inspect(out1, "est")$theta - inspect(out2, "est")$theta

     Q77 Q84 Q170 Q196 Q44 Q63 Q76 Q98
Q77    0
Q84    0   0
Q170   0   0    0
Q196   0   0    0    0
Q44    0   0    0    0   0
Q63    0   0    0    0   0   0
Q76    0   0    0    0   0   0   0
Q98    0   0    0    0   0   0   0   0
```